MIL-W-5216 lD(AL)
23 June 1989
SUPERSEDING
MIL-W-52161C(AL)
13 June 1983

MILITARY SPECIFICATION

WELDING SET, ARC, INERT GAS, SHIELDED:

CONSUMABLE WIRE ELECTRODE

This specification is approved for use by the Armament Research, Development and Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of arc welding sets consisting of hand held welding guns with built-in aluminum wire electrode feeding mechanisms that accommodate one pound size wirspools, and that are furnished with a welding current, inert gas flow, and wire feeder control box. The welding sets are for use with DC power supplies and inert gas shielded arc welding process.

1.2 Classification. The welding sets shall be of the following types, as specified (see 6.2.1).

Type I - With contactor assembly, for use with constant current and constant voltage DC welding power supplies and for use with 3/64 inch diameter electrode wire on a one pound spool.

Type II - Without contractor assembly, for use with constant voltage DC welding power supplies, and for use with 0.030 and 3/64 inch diameter wire electrodes on one pound spools.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitatim (see 6.2. 1).

SPECIFICATIONS

FEDERAL

J-C-580	-	Cord, Flexible, and Wire, Flexible (Electrical, 0 to 600 Volt Service)
W-C-596	-	Connector, Plug, Receptacle and Cable Outlet, Electrical Power
QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet, General Specification for

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture and Fungus Resistant (For Treatment of Commnication, Electronic, and Associated Equipment)
MIL-P-514	-	Plate, Identification Instruction and Marking Blank
MIL-T-704	-	Treatment and Painting of Materiel
MIL-C-4150	-	Case, Transit and Storage, Waterproof and Water-Vapor-proof
MIL-G12044	-	Chest, Plywood
MIL-W-45562	-	Welding and Soldering Equipment, Supplies, and Accessories; Packaging of

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards For Federal Services

MILITARY

MILSD-130	-	Identification Marking of US Military Property
MIL-SD-889	-	Dissimilar Metals
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

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MS3100 - Connector, Receptacle, Electrical, Wall Mounting Solder Contacts, AN Type

MS3106 - Connector, Plug, Electric, Straight, Solder Contacts, AN Type

DRAWINGS

AL

D13205E5433 - Welder Schematic Diagram

(Unless otherwise indicated, copies of federal and military sepcifications, standards, and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120 -5099.)

2.2 NonGovernment publicaticms. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of-the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (See 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AWS A5.10 - Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes

(Application for copies should be addressed to the e American National Standards Institute, Inc., 1430 Broadway, New York, NY 100I8.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A569 - Steel, Carbon (0.15 Maximum Percent), Hot Rolled, Sheet and Strip, Commercial Quality

ASTM A684 - Steel, Carbon, Strip, Cold Rolled, Soft, Untempered Spring Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

COMPRESSED GAS ASSOCIATION (CGA)

CGA Pamphlet E-4 - Standard for Gas Regulators for Welding and Cutting

(Application for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Alexandria, VA 22202.)

(Non-Government standards and other publications are nomally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more complete welding machines for first article inspection (see 4.3 and 6.2.1).

3.2 Design and description. The welding set shall be new and of current standard commercial design meeting the requirements of this specification. A welding set shall consist of an air-cooled, semi-automatic, wire feed, pull type welding gun; a portable welding control unit; and all other equipment and accessories necessary for standard arc welding processes.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor from standard commercial materials and shall be adequate for the use intended.

3.3.1 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Sheet and strip steel. Sheet and strip steel shall conform to ASTM A 569 or ASTM A 684 as applicable.

3.3.3 Sheet aluminum. Sheet aluminum shall conform to QQ-A-250, alloy optional.

3.3.4 Cable and cord. Cable and cord used between the welding power source and the control box shall conform to type SO of J-C-580. Cable and cord used between the control box and the hand held gun shall conform to either type SO or type SJO of J-C-580. Cable and cord shall be supplied in lengths that permit the operation of the welding gun at least 50 feet away from the power supply and at least 40 feet away from the welding control unit.

3.3.5 Inert gas hose. The inert gas hose shall be made of heat and oil resistant, vinyl-resin or synthetic rubber. The hose shall not be hydrostatically tested.

3.3.6 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.7 Threads. The design, manufacture, and configuration of threads shall conform FED-STD-H28 and the applicable detailed standards referenced therein.

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3.3.8 Funguss control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MILT-152 and MIL-V-173.

3.4 Operating envronmental conditions. The welding sets shall operate, as specified herein, in ambient temperatures from -25 degrees Fahrenheit (F.) (any humidity) to +110 degrees F., with a reative humidity of 90 percent.

3.4.1 Storage environmental conditions. The welding sets shall not be damaged physically or have their operating capabilities diminished by being stored at temperatures as low as -30 degrees F. (any humidity) and as high as +160 degrees F. with relative humidity of 90 percent and all temperatures between -30 degrees F. and +160 degrees F. any humidity),

3.5 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement. Parts subject to wear shall be accessible for adjustment, replacement, or repair.

3.6 Rough handling. The welding set, while packed in its permanent storage box (not the shipping container) shall be capable of withstanding, without damage, the cumulative effect of shocks resulting from being dropped ten successive times from a height of at least six inches to a solid floor.

3.7 operation. When connected to a DC power supply, the welding set shall have specific welding characteristics, so that an experienced welding operator can strike and maintain a stable arc and weld, within the applicable ranges associated with the material, and the diameter of an electrode used at currents not exceeding 250 amperes. Electrodes shall conform to ANSI/AWS A5.10.

3.8 Endurance. The welding set shall withstand without damage the effects of momentarily terminating the welding arc by pulling the gun away from the work piece without releasing the trigger switch.

3.9 Welding gun. The welding gun shall be hand held, semiautomatic, air-cooled, with a pull type wire feed mechanism built into the gun and a housing for a one pound wire spool built into the gun. The gun shall incorporate mechanisms and controls for feeding the win from the one pound spool during welding, inching the wire when not welding and starting and stopping the flow of shielding gases. The gun shall be constructed so that it can be held and operated in all welding positions with one hand. The gas nozzle of the gun shall direct shielding gases in such a way that the shielding gases block out any contaminating atmosphere from the weld puddle area. The gas nozzle and the electrode wire guide tube shall be removable. The parts used to enclose the wire feed mechanism and the one pound wire spool housing shall be sufficiently strong to withstand transport, set-up and normal shop use and handling.

3.9.1 Electrode wim spool housing. The welding gun electrode wire spool housing shall accommodate a standard one pound spool of aluminum electrode wire. The housing shall protect the wire from dust, spatter, dirt, and rain. The housing shall have a removable cover for access to the spool of electrode wire. The cover shall be held in the closed position by quick disconnect fastener(s) Permanently fixed to the housing. A locking device shall be provided in the housing for holding the spool of electrode wire in place. A device shall be provided to prevent the electrode or wire from unwinding faster than the feed of the gun.

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3.9.2 Triggerswitch. A grip type trigger switch shall be provided in the welding gun handle to activate the motor of the electrode wire feed mechanism and concurrently permit the flow of inert gas and welding current to the gun.

3.9.3 Electrode inching switch. A switch shall be provided in the welding gun handle for inching the electrode wire without permitting the flow of the welding current and inert gas to the welding gun.

3.9.4 Electrode wire feed mechanism. A pull type, motor driven, electrode wire feed mechanism shall be provided in the welding gun. The mechanism shall consist of a set of wire feed rolls for 3/64 (0.0469) inch aluminum wire, a drive motor with appropriately designed torque output, and supporting parts. The wire feed roll shall have a knurled surface. Material for the wire feed roll shall be fully hardened tool steel with a hardness of Rockwell C 58 to 64, or case hardened steel with a minimum hardness of Rockwell superficial 15N88 and a minimum case depth of 0.010 inch. The wire feed mechanism shall be fully enclosed and shall have a removable cover providing access to all parts of the wire feeder mechanim and to the wire spool.

3.10 Portable welding control unit. The portable welding control unit shall consist of all electrical, electronic, and mechanical components and circuitry necessary to control the wire feed mechanism. The welding control unit enclosure shall be constructed of sheet metal not less than 0.048 inch thick. Total weight of the complete unit shall not exceed 40 pounds and overall size (height, width, and length) shall not exceed 2880 cubic inches, excluding carrying handle. Fuses or circuit breakers shall be provided for protection of the drive motor and control circuits.

3.10.1 Wire feed speed control. A wire feed speed control shall be provided for regulating the electrode wire feed speed to permit welding with the type and size of electrode wire specified herein (see 3.9.4 and 3.12.5). The Wire feed speed control may be located at either the control unit or at the welding gun. Means shall be provided to maintain constant welding wire feed speed, irrespective of load variations imposed on the drive motor.

3.10.2 Gas valve. A gas valve shall be provided to start and stop the flow of shielding gas. The valve shall be controlled by the trigger switch on the gun. The control mechanism shall be adjusted to provide gas flow when the trigger is partially depressed without energizing the welding circuit.

3.11 Hose lengths. Inert gas hose shall be provided in lengths that permit operating of the welding gun-50 feet away from the gas supply and 40 feet away from the welding control unit. Inside diameter of the gas hose shall be not less than 3/16 inch. The insulation around the welding conductor may be used to conduct inert gases providing it fits loosely enough to permit the rate of flow required to do the heaviest welding and is as puncture proof as a separate inert gas hose.

3.12 Type I welding set. The type I welding set skll operate with both constant current and constant voltage power supplies, and shall accommodate a 3/64 inch diameter aluminum electrode. The welding set control shall be designed to operate in conjunctim with either a constant current (cc) or constant voltage (cv) power supply, and shall be equipped to change from cc to cv and cv to cc without the necessity for electrical disconnections and or re-connections. If a switch is required to accommodate this purpose, the switch shall be externally accessible, and skall be protected to prevent accidentally changing from cv to cc and from cc to cv.

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3.12.1 Welding current contactor. A welding current contactor shall be provided in each welding current control unit. The contactor shall start and stop the welding currents within the operating range of the welding set.

3.12.2 Input power. The welding set shall be designed to operate on 115 volts AC, 60 Hertz, and 115 volt DC input power. At least 10 feet of a three conductor, flexible, electrical supply cord shall be provided with each welding control unit. The power supply cord shall terminate in a three grounding type male connector plug, style D series, conforming to W-C-596.

3.12.3 Gas regulator and flowmeter. A two-stage gas regulator complete with flowmeter shall be provided for controlling and measuring the cylinder pressure, and the flow of inert gases to the gun. The gas regulators and flowmeters shall conform to the Compressed Gas Association Standard for Gas Regulators for Welding and Cutting, CGA Pamphlet E-4-77, and the following requirements.

3.12.3.1 Design. Regulators shall reduce the gas cylinder pressures indicated and deliver the gases for which they are designed at any desired regulated pressure or flow without vibrating or humming.

3.12.3.2 Valve construction. Seating shall be accomplished between a metallic and a nonmetallic material. The relatively soft seating member and the mating parts shall be held in alignment with each other in such a way as to maintain the sensitivity of the regulator.

3.12.3.3 Flowmeters. Regulators shall be equipped with, and shall discharge through tapered tube with ball type flowmeters, which are calibrated to indicate directly the flew of argon and helium, respectively, in cubic feet per hour (cfh). Minimun range of indication shall be 10 to 50 cfh argon, and 20 to 150 cfh for helium. Flow indication in these ranges shall be accurate within 10 percent of the true flow. Design of flowmeters shall afford the tubes protection against accidental breakage. Flowmeters may be either single or dual range type. If dual range, the scales shall incorporate an indicating method that positively identifies the correct scale to read.

3.12.3.4 Marking. Each regulator shall be clearly and permanently marked or labeled with the manufacturer's name or trademark, model designation, operating inlet pressure, and type of gas it is to be used with.

3.12.4 Storage chest. A chest, conforming to MIL-C-4150 or MIL-C-12044, shall be furnished with each set. The chest shall be of adequate size to store the welding set, accessories. cables, and maintenance tools. When conforming to MIL-C-4150, the chest shall be Type 1 (waterproof), Style 1 (transit case), Class A (gross weight up to 150 lbs), and Variety HC (hinged closure). When conforming to MIL-C-12044, the wood shall be group B plywood of 5/8 inch nominal thickness, exterior type, and finished on one side. Provisions shall be made to hold the articles securely in place in order to prevent displacement or damage during transit. The storage chest shall be marked with the National Stock Number Military Specification Number, and item nomenclature of the welding set in accordance with MIL-STD-130. When provided in accordance with MIL-C-4150 the storage chest shall conform to the following requirements:

a. Four handles with no handles on the lid.

b. No lifting rings.

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c. No automatic pressure-vacuum relief valve.

d. No humidity indicator.

e. Cushioning and shock mount ing shall be provided.

f. Mark with end item nomenclature and National Stock Number (the welding set number).

g. Name plate shall be required.

h. Finish in accordance with the color and finish requirements of MIL-C-4150, with the color to be forest green.

i. Clasps as specified in MIL-C-4150.

3.12.5 Accessories. The following accessories shall be furnished with each Type I set:

Quartity	Item
20	- One pound spools of wire electrodes, 3/64 inch aluminum wire, ANSI/AWS A5.10
36	- Guide tubes.
5	- Gas nozzles.
2	- Extra line fuses of each type used.
20	- Extra drive titer fuses.
1	- Extra set of wire feed rolls.

3.13 Type II welding set. The Type II welding set shall operate with constant voltage power supplies, and shall accommodate both 0.030 inch and 3/64 inch diameter electrodes. Input power shall be 115 volts AC, 60 Hertz.

3.13.1 Input power and contactor connection cord. At least 10 feet of five conductor, flexible, electrical supply cord shall be provided with each welding control unit. The cord shall terminate in a connector conforming to MS3106-F18-11P. When mated with connector MS3100-R18-11S (identified as J3 on welder schematic drawing D13205E5433), the cord skill transmit input power to the control unit and a signal to the welding machine contactor. The connector shall be wired as follows:

a. Input power - Connect to prongs C and D.

b. Contactor signal - Connect to prongs A and B if 115 volt AC or DC signal is furnished. Connect to prongs A and D with a jumper wire between B and C if a switch closing circuit signal is furnished.

Co Ground - Connect to prong E.

3.13.2 Accessories. The following accessories shall be furnished with each Type II set:

Quantity	Item
2	- Extra line fuses of each type used.
20	- Extra drive motor fuses.
1	- Extra set of wire-feed rolls.

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5	- Gas nozzles.
36	- Guide tubes for 0.030 inch wire.
36	- Guide tubes for 3/64 inch wire.
1	- Set of guide bushings for each size wire.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with standard means for lubrication.

3.15 Product identification. The welding set shall be permanently identified in accordance with MIL-STD-130 including, unless otherwise specified (see 6.2.1), the National Stock Number and the contract number. The marking shall be applied to the welding set on identificati0n plates conforming to MIL-P-514, Type I, Style 1, Composition A (Class 2), or Composition C (of Type I), Grade A, Class 1 material. The plates shall be attached by screws, bolts, or rivets in a conspicuous but protected location

3.16 Instruction plates. Each welding set shall be quipped with instruction plates, including warnings and cautions, which describe any special or important procedures to be followed in the operation of the welding set. Instruction plates shall conform to MIL-P-514, Type III, Composition A (Class 2), or Composition C (of Type I), Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous but protected location.

3.17 Paintirg Unless otherwise specified (see 6.2.1), portions of the welding sets which are normally painted shall be cleaned, treated, and painted in accordance with standard commercial practice. At least one coat of primer and one coat of enamel, or equivalent, shall be applied. The storage chest (see 3.12.4), when furnished in accordance with MIL-C-12044, shall be treated and painted forest green as specified in MIL-T-704, Type A.

3.18 Workmanship. Standards of workmanship shall assure that the welding sets shall have the stability and efficient operating characteristics found in standard commercial items and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 50 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requiements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submissim of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspect (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of a. through-c. below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).

b. Tests (see 4.6.1, 4.6.2, 4.6.3, and 4.6.6).

c. Packaging inspection (see 4.7).

4.5 Product examination Visually and manually examine the welding set to determine conformance with the requirements of 3.2 through 3.5 and 3.9 through 3.18. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the welding set and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Test conditions. The welding set shall be connected to a suitable DC power supply to perform the tests specified herein. Actual welding operations shall be performed using both constant current and constant voltage power for Type I sets, and constant voltage power for Type II sets. Welds required in 4.6.2 shall be made with 3/64 inch diameter wire for Type I sets, and with 0.030 inch and 3/64 inch diameter wire for Type II sets. Electrodes shall conform to ANSI/AWS A5.10 with the work piece being made of compatible material.

4.6.2 Welding characteristics. Using at least two current settings nomally associated with the diameter of wire used, a series of single-pass and mulit-pass fillet joint welds shall be made. Each weld shall be at least 12 inches long. Loss of the welding arc at a frequency greater than once per minute, or nonconformance to 3.7 and 3.10.1, shall constitute failure.

4.6.3 Mechanical properties. Using a 3/64 inch diameter electrode, a weldment shall be fabricated and tested in accordance with ANSI/AWS A5.10 including the follwing:

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a. Radiography.

b. Face and root bend test.

c. Transverse tensile test.

Inability of the specimens to meet the applicable test requirements shall constitute failure.

4.6.4 Endurance. While the welding sit is operating at not less than 200 amps, the are shall be terminated by pulling the gun away from the work piece rather than by releasing the trigger switch. Repeat this procedure for 25 cycles. Evidence of damage to the welding set, or nonconformance to 3.7 and 3.8, shall constitute failure.

4.6.5 Rough handling. The welding control unit shall be dropped ten times to a solid floor from a height of six inches. Evidence of damage during and after the ten drops shall constitute failure of the test (see 3.6).

4.6.6 Operational. The welding set shall be energized, and all controls and assemblies operated to simulate actual welding conditions. The contractor's final assembly test shall be acceptable provided the test and recorded performance data indicate compliance with the applicable requirements. Failure of any component or assembly shall constitute failure to pass the test.

4.6.7 Operating environment. The welding set shall be placed in an ambient temperature of +105 to +110 degrees F. The test in 4.6.6 shall be performed after the unit has stabilized at this temperature. The welding set shall-then be placed in an ambient temperature of -20 to -25 degrees F. and the test in 4.6.6 repeated after the unit has stabilized at this temperature. Inability of the unit to meet the test requirements at specified temperatures shall constitute failure of this test (see 3.4).

4.6.8 Storage environment. The welding set shall be placed in an ambient temperature of +155 to +160 degrees F. for a period of 72 hours. At the end of this time the welding set shall be placed in an ambient temperature of -25 to -30 degrees F for 72 hours. At the end of this period, the welding set shall be observed for damage and the test in 4.6.6 shall be performed. Any damage observed or failure of the unit to meet test requirements shall constitute failure of the test (see 3.4.1).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level A in accordance with MIL-W-45562 or Level C in accordance with MIL-STD-ll90, as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The welding sets covered by this specification are intended to be used for semi-automatic, arc welding of aluminum.

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6.2 Ordering data.

6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type of set required (see 1.2).

c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).

d. First article inspection, when required (see 3.1).

e. Identification marking, if different (see 3.15).

f. Painting, if different (see 3.17).

g. First article inspection, indifferent (see 4.3).

h. Quality conformance inspection, if different (see 4.4).

i. Selection of required level of preservation, packing, and marking (see 5.1).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Contract data requirements. Required technical data such as operator's manuals. parts lists, and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on DD Form 1423- incorporated in the contract.

6.5 Subject term (key word) listing.

Inert gas
Wirespool
Regulator
Flowmeter

6.6 Sampling. Sampling for quality conformance inspection should be performed as follows. Lots should be accepted with zero defects and rejected if one or more defects are found.

INSPECTION OR TEST	LOT OR BATCH SIZE	NUMBER RANDOMLY SELECTED FOR INSPECTION
Product examination (4.5)	----	100 percent inspection
Welding characteristics (4.6.2)	1-150	8
Mechanical properties (4.6.3)	1-150	8
Operational (4.6.6)	----	Test each item
Packaging inspection (4.7)	1-150	5

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Army - AL

User activity:
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Preparing activity:
Army-AL

Project No. 3431-A194

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-W-52161, Revision D (AL), Welding Set, Arc, Inert Gas, Consumable Wire Electrode Shielded:

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED.

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